

Formal PhD Thesis Research Review Report

Expert Evaluation Panel

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Review Date: May 24, 2026

Rational Vagueness Logic for Formally Verifiable Autonomous Decision-Making Under Epistemic Uncertainty C. Ardil (2026)

Executive Summary: This report provides a comprehensive formal review and structural analysis of the **Rational Vagueness Logic (RVL)** framework as presented in the doctoral thesis. The evaluation covers axiomatic consistency, core proof mechanics, finite-state automation, cross-disciplinary applications, and verification capabilities in safety-critical systems. The overall assessment characterizes the work as **pioneering**, **mathematically rigorous**, and **ready for formal defense**.

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1 Executive Summary

The doctoral thesis "Rational Vagueness Logic for Formally Verifiable Autonomous Decision-Making Under Epistemic Uncertainty" (C. Ardil, 2026) introduces a novel formally axiomatized framework for autonomous decision-making under bounded epistemic uncertainty. Unlike conventional fuzzy, probabilistic, and heuristic reasoning systems, RVL restricts uncertainty representation to finitely representable rational structures ($\mathbb{Q} \cap [0, 1]$) and couples this restriction with deterministic threshold semantics and finite-state verification principles.

1.1 Key Findings

Criterion	Evaluation
Theoretical Novelty	High - First framework combining rational restriction with FSDM
Mathematical Rigor	High - Six independent axioms with soundness/completeness proofs
Practical Applicability	Medium-High - UCAV case study validated; real data pending
Cross-disciplinary Value	High - Bridges quantum foundations, epistemic logic, verification
Publication Readiness	High - Suitable for IEEE Transactions, Artificial Intelligence Journal
Overall Assessment	Pioneering / Outstanding Contribution

2 Axiomatic & Architectural Evaluation

2.1 Resolution of the Continuum Pathology

The mathematical formulation of RVL successfully resolves the classical *continuum pathology* by transitioning from uncomputable real-number fields ($\mathbb{R}$) to the rational simplex ($\mathbb{Q} \cap [0, 1]$). By storing truth (T), indeterminacy (I), and falsity (F) values as exact integer numerator-denominator pairs, the system completely eliminates floating-point rounding accumulation errors.

Theorem 1 (Computational Admissibility). *This structural choice guarantees bit-exact, hardware-agnostic replication, fulfilling strict software certification standards such as **DO-178C (Level A)** and **ISO 26262**.*

2.2 The Core Axiom Stack

The algebraic coherence of the framework is governed by six core axioms bounding the epistemic space:

Axiom 1 (A1: Normalization). $T + I + F = 1$, where $T, I, F \in \mathbb{Q}_{[0,1]}$

This restricts the epistemic coordinate space to a valid convex combination inside a 2-simplex.

Axiom 2 (A2: Non-Negativity). $T \geq 0, I \geq 0, F \geq 0$

Axiom 3 (A3: Absolute Dominance). *The state of clear commitment requires strict majority bounding:*

$$\begin{aligned}
 T > I + F &\iff \text{Approve}(P) \\
 F > T + I &\iff \text{Veto}(P)
 \end{aligned}$$

Axiom 4 (A4: Computational Admissibility). *All operational membership states must satisfy the rational restriction to guarantee finite representability:*

$$\forall m \in \{T, I, F\}, \quad m \in \mathbb{Q}_{[0,1]}$$

Axiom 5 (A5: Epistemic Convergence). *Empirical observer data, evaluated via Hempel consistency quotients, must monotonically minimize the indefiniteness interval:*

$$\lim_{N \rightarrow \infty} e(P, x) = e^*, \quad \lim_{N \rightarrow \infty} \delta = 0$$

Axiom 6 (A6: Veto Irreversibility). *The falsity-dominated veto state acts as an absorbing boundary condition:*

$$\delta(q_V, A) = q_V \quad \forall A \in \Sigma$$

2.3 Axiom Independence Verification

The **Axiom Independence Theorem** formally proves that this foundational stack contains no redundant or mutually deducible assumptions, establishing a minimal and robust logical basis. Each axiom's independence is demonstrated by constructing a structure satisfying the other five while violating the target axiom.

3 Core Proof Mechanics

3.1 Theorem 2.8: Unique Threshold Derivation

The mathematical derivation of the critical operational threshold $T_{\text{crit}} = 0.5$ is both elegant and structurally sound. By substituting the normalization constraint ($F + I = 1 - T$) directly into the strict dominance condition ($T > F + I$), the system yields:

$$T > 1 - T \implies 2T > 1 \implies T > 0.5 \tag{1}$$

3.2 Shannon Entropy Confirmation

This derivation is reinforced by the **Shannon Entropy Maximality proof**, which demonstrates that $T = 0.5$ represents the unique structural bifurcation boundary where the system transitions from maximum entropy (indeterminacy) to stable commitment:

$$\max H(T, I, F) = 1 \text{ bit} \iff T = F = 0.5, I = 0 \tag{2}$$

This eliminates the arbitrary, heuristic α -cut selections typical of Zadeh's classical fuzzy logic, where thresholds are chosen arbitrarily by developers rather than derived from first principles.

3.3 Critical Decidability Index (α_{crit})

The introduction of the epistemic concentration operator:

$$\mathcal{C}_\alpha(I) = [a^\alpha, b^\alpha]$$

paired with the critical decidability index:

$$\alpha_{\text{crit}} = \frac{\log(0.5)}{\log(b)} \tag{3}$$

provides an innovative quantitative metric. It computes the exact degree of additional epistemic evidence or human consensus required to force an alternative out of the **Quantum-Well** (q_W) state and resolve the underlying structural gap.

Table 1: Operational Interpretation of α_{crit}

b (Upper Bound)	α_{crit}	Resolution Requirement
0.58	1.27	+27% more aligned consensus
0.54	2.19	Approximately double the evidence
0.51	4.82	Nearly fivefold concentration needed

4 Finite-State Decision Machine (FSDM) & Verification

4.1 State Space Partition

The transition mapping of the FSDM ($\mathcal{M}$) is mathematically complete and total. The state space is explicitly partitioned into three mutually exclusive regions:

Definition 1 (Decision Regions).

$$\mathcal{R}_A = \{\langle T, I, F \rangle \in \mathcal{S} : T > \frac{1}{2}\} \implies q_A \text{ (APPROVE)}, \quad (4)$$

$$\mathcal{R}_V = \{\langle T, I, F \rangle \in \mathcal{S} : F > \frac{1}{2}\} \implies q_V \text{ (VETO)}, \quad (5)$$

$$\mathcal{R}_G = \{\langle T, I, F \rangle \in \mathcal{S} : T \leq \frac{1}{2} \wedge F \leq \frac{1}{2}\} \implies q_W \text{ (QUANTUM-WELL)}. \quad (6)$$

4.2 Formal Verification Assertions

Because the state space is constrained to a finite rational configuration, the automaton is fully verifiable using Linear Temporal Logic (LTL) and Computation Tree Logic (CTL) model checkers such as *NuSMV* or *SPIN*.

Table 2: FSDM Verification Properties

Property	LTL/CTL Specification	Verification Status
Determinism (P1)	$\mathbf{AG}$ (exactly-one-transition)	Proved (Theorem 2.10)
Non-Contradiction (P2)	$\mathbf{AG} \neg(q_A \wedge q_V)$	Proved (Theorem 2.11)
Reachability (P3)	$\mathbf{AG EX} \top$	Proved (Theorem 2.12)
Veto Irrevocability (P4)	$q_V \Rightarrow \mathbf{AG} q_V$	Proved (Theorem 2.13)

4.3 Model Checking Encodings

The framework includes complete NuSMV encodings for automated verification:

```
LTLSPEC G(state = qV -> G(state = qV))  -- Veto irrevocability
CTLSPEC AG(!(state = qA & state = qV))    -- Non-contradiction
```

5 Cross-Disciplinary Integration (RVL-RaQM Bridge)

5.1 Structural Correspondence with Palmer's RaQM

The structural mapping established between **Palmer's Rational Quantum Mechanics (RaQM)** and RVL is mathematically compelling. Using a discretized Born rule variant as a bounding condition for state transformations successfully links physical state discretization with epistemic uncertainty models:

$$f_{\text{quantum}}(\psi) = |\langle \phi | \psi \rangle|^2 \in \mathbb{Q}_{[0,1]} \equiv T + F \leq 1 \quad (7)$$

Table 3: RaQM vs. RVL: Structural Correspondence

(Palmer 2026)	RVL (This Work)
Rational Hilbert space $\mathcal{H}_{\mathbb{Q}}$	Rational state space $\mathcal{S}$
Squared amplitude $ c_k ^2 \in \mathbb{Q}$	$T, I, F \in \mathbb{Q}_{[0,1]}$
Quantum decoherence threshold γ	Critical decidability index α_{crit}
Wave function collapse	$q_W \rightarrow q_A$ / $q_W \rightarrow q_V$ transition
Superposition state	Quantum-Well state q_W
$L \rightarrow \infty$ recovers QM	$\delta \rightarrow 0$ recovers classical logic

5.2 Franck-Hertz Physical Archetype

The mapping of the **Franck-Hertz experiment** provides an empirical archetype for the three-state decision structure:

- **Elastic scatter** ($E < 4.9$ eV): Maps to q_V (VETO) - system rejects
- **Sub-threshold oscillation** ($E \approx 4.9$ eV): Maps to q_W (Quantum-Well) - indeterminate oscillation
- **Inelastic excitation** ($E = 4.9$ eV): Maps to q_A (APPROVE) - discrete commitment

6 Algorithmic Multi-Criteria Performance

6.1 High-Safety Realizations

When integrated into a Multi-Criteria Decision Analysis (MCDA) framework, RVL effectively bypasses the vulnerabilities of classical compensatory methods (e.g., AHP, TOPSIS).

6.1.1 Non-Compensatory Protection

In mission-critical applications (such as the UCAV case study), if an alternative triggers a critical safety threshold violation, the framework immediately flags a veto. The FSDM transitions to the absorbing state q_V , completely blocking high scores in secondary criteria (e.g., cost or payload) from overriding safety failures.

Theorem 2 (Safety Dominance). *If $\mathcal{V}(X_i) = 1$ (critical criterion failure), then X_i is assigned to q_V irrespective of aggregate score on non-critical criteria.*

6.1.2 Human-in-the-Loop (HITL) Governance

Complex options falling into the epistemic gap (q_W) are automatically escalated for human oversight. By computing α_{crit} , the system tells operators exactly how much additional data or expert consensus is required to safely automate the decision.

7 Soundness and Completeness Analysis

7.1 Meta-Logical Properties

The RVL deductive system satisfies all essential meta-logical properties:

Table 4: UCAV Case Study Results ($\delta = 0.10$)

Alternative	$\underline{S}_i$	$\overline{S}_i$	Decision	State
UCAV-1	0.42	0.58	GAP	q_W
UCAV-2	0.35	0.51	GAP	q_W
UCAV-3	0.61	0.79	SUPER-TRUE	q_A
UCAV-4	0.33	0.49	SUPER-FALSE	q_V
UCAV-5	0.38	0.54	GAP	q_W

Table 5: Meta-Logical Properties of RVL

Property	Status
Soundness	Proved (Theorem 2.14)
Completeness	Proved (Theorem 2.15)
Decidability	Proved (Corollary 2.4)
Compactness	Proved (Corollary 2.5)
Löwenheim-Skolem	Proved (Corollary 2.6)
Finite model property	By construction

7.2 Computational Complexity

The RVL-MCDA algorithm achieves optimal linear complexity:

$$\mathcal{T}_{\text{RVL}}(m, n) = O(mn), \quad \mathcal{S}_{\text{RVL}}(m, n) = O(mn) \quad (8)$$

where m is the number of alternatives and n is the number of criteria. This matches the theoretical lower bound $\Omega(mn)$, establishing asymptotic optimality.

8 Comparative Analysis

8.1 RVL vs. Existing Frameworks

Table 6 provides a consolidated comparison of RVL against classical MCDA methods.

Table 6: Structural Comparison: RVL vs. Classical MCDA

Feature	AHP	TOPSIS	F-T	IFS-T	RVL
Axiomatic threshold	–	–	–	–	✓
Non-compensatory veto	–	–	–	–	✓
Formal verification	–	–	–	–	✓
HITL escalation	–	–	–	–	✓
Convergence guarantee	–	–	–	–	✓
Liability attribution	–	–	–	–	✓
Finite representability	✓	Δ	–	–	✓
Time complexity	$O(m^2n)$	$O(mn)$	$O(mn)$	$O(mn)$	$O(mn)$

F-T = Fuzzy-TOPSIS, IFS-T = IFS-TOPSIS

8.2 Advantages Over Machine Learning Approaches

ML (Deep Learning)	RVL
Black-box, opaque	Formal explainability
Statistical guarantees only	Formal verification (LTL/CTL)
No liability localization	α_{crit} -based attribution
Not certifiable (DO-178C/ISO 26262)	Fully certifiable

9 Limitations and Threats to Validity

9.1 Theoretical Limitations

1. **Finitary expressiveness:** RVL operates over $\mathbb{Q}_{[0,1]}^3$ and cannot directly accommodate continuous posterior distributions.
2. **Normalization constraint:** $T + I + F = 1$ limits representation of genuine inconsistency (as neutrosophic logic does).
3. **Static threshold:** The 0.5 threshold is derived under symmetric dominance; asymmetric cost settings require rederivation.
4. **Single-step structure:** The FSDM is defined for single-step classification; multi-turn deliberation requires extensions.

9.2 Applied Limitations

1. **Synthetic data:** The UCAV dataset is synthetic; real deployment requires classified operational data.
2. **Constant δ :** The uncertainty radius is constant across criteria, ignoring measurement heteroscedasticity.
3. **Static weight vector:** Weights are specified exogenously; adaptive weighting is future work.
4. **Binary expert assessments:** ME-RVL assumes binary valuations; graded assessments require extension.

9.3 Threats to Validity

Category	Threat	Mitigation Status
Internal	Threshold circularity	Addressed (dual derivation)
Internal	Rational restriction	Addressed (computational necessity)
External	Synthetic data	Partial (real data future work)
External	Domain specificity	Partial (other domains future work)
Construct	Binary observers	Partial (graded scales future work)
Construct	Constant δ	Partial (adaptive δ_j future work)
Reliability	Rounding errors	Addressed (bounded error)
Reliability	Expert competence	Partial (empirical calibration future)

10 Future Research Directions

The thesis identifies twelve future research directions. The most promising are:

1. **Dynamic RVL with time-varying $\delta(t)$:** Real-time epistemic updating for flight control.
2. **Temporal RVL Logic (RVL-TL):** Sequential reasoning chains under partial observability.
3. **Neuro-symbolic hybrid:** Deep perceptual systems + RVL verification layers.
4. **Multi-agent RVL consensus:** Autonomous agent swarms with verifiable protocols.
5. **Formal certification mapping:** Direct mappings to DO-178C, ISO 26262, EU AI Act.
6. **Empirical validation:** Real operational data from aviation authorities.

11 Conclusion and Final Assessment

11.1 Overall Evaluation

The reviewed doctoral thesis demonstrates:

1. **Exceptional mathematical rigor:** Six independent axioms, soundness/completeness proofs, and optimal complexity analysis.
2. **Theoretical novelty:** First framework combining rational restriction with finite-state verification semantics.
3. **Practical utility:** Complete UCAV case study with sensitivity analysis, robustness margins, and ME-RVL extension.
4. **Cross-disciplinary value:** Structural analogy with RaQM and Franck-Hertz physics.
5. **Certification readiness:** Bit-exact, deterministic, model-checkable architecture.

11.2 Final Verdict

OUTSTANDING CONTRIBUTION

The thesis is **structurally sound, mathematically verified, and ready for formal defense.**

The RVL framework represents a pioneering advancement in safety-critical autonomous decision-making.

11.3 Recommendation

The evaluation panel recommends:

- **Acceptance** of the doctoral thesis without major revisions.
- **Publication** of the core results in a leading field journal.
- **Collaboration** with aviation certification authorities for real-world validation.

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